

Name: Stream:

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P525/1
CHEMISTRY
Paper 1
(Theory)
April, 2026
2¾ Hours

MENGO SENIOR SCHOOL
Uganda Advanced Certificate of Education
END OF TERM I ASSESSMENT 2026
S.5 CHEMISTRY
PAPER 1
2 HOURS 45 MINUTES

INSTRUCTIONS TO CANDIDATES:

- This paper consists of **four** examination items.
- Attempt **all items** in Section A and Section B.
- Responses to items in Section A are to be written in the spaces provided while in Section B part I (Inorganic) and part II (Organic) **must** be written on separate answer booklets.
- Use **blue** or **black** ink.
- All working must be clearly shown.
- Graph paper is provided.
- A Periodic Table is provided.
- Mathematical tables and silent non- programmable calculators may be used.
- You are advised to carefully read each item and the **plan** appropriately before starting.

For Scorer's Use Only

Item	1	2	3	4
Weighted Score				
Scorer's Initials				
TOTAL SCORE				

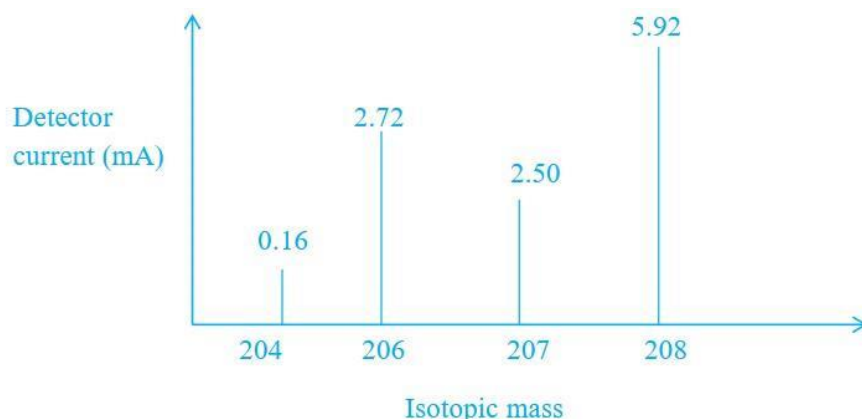
SECTION A

*Attempt **all** items from this section*

Item 1:

(a) The Uganda National bureau of standards (UNBS) received complaints about metal contamination in several imported fruit juices after consumers reported symptoms of headaches, stomach pain and dizziness. The agency suspected that the juice might contain a toxic metal but the exact metal was unknown. When the juice sample was taken and analyzed, the suspected metal in juice had mass spectrometry analysis data as shown in support material below.

During analysis the sample was ionized and then sorted into different ions which were detected.



Task; As a chemistry student with knowledge about spectrometry;

(i) Describe how the mass spectrometer **ionizes** and then **separates** the ions in order to produce the results the analytical agency could have obtained. (05 scores)

[illegible]

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- (01 score)**

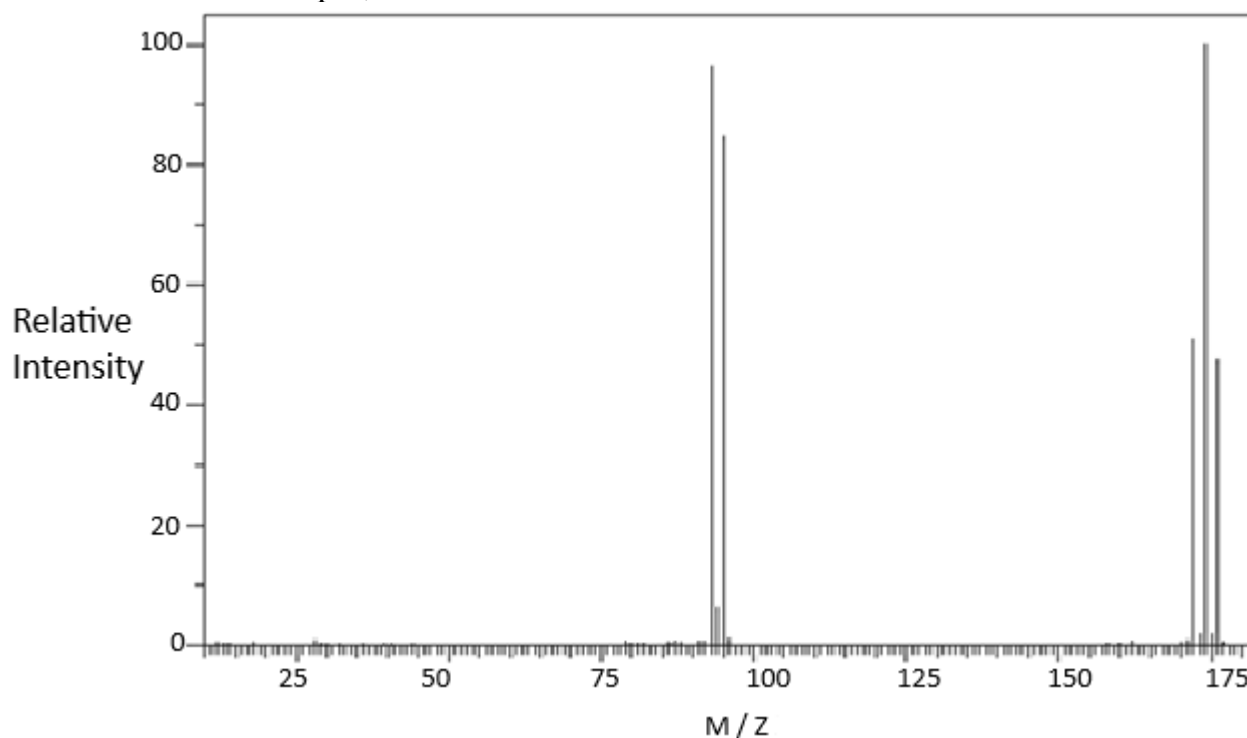
(04 scores)

(02 scores)

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(iv) State one advantage of using the mass spectrometer to determine the toxic metal in the juice.

(01 score)

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(b) An analysis of **dibromomethane, CH_2Br_2** using a mass spectrometer shows molecular peaks of mass-charge ratio at 79, 81, 172, 174 and 176. The relative atomic mass of bromine is 79.9. Bromine has two isotopes, Bromine-79 and bromine-81.



This information confuses Peter a new S.5 chemistry student and he needs your assistance;

Task: As a chemistry learner, guide Peter about;

(i) the species of ions responsible for the three lines at peaks 172, 174 and 176.

(03 scores)

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(ii) how to calculate the relative percentage abundance of each bromine isotope.

(03 scores)

(iii) one other special application of spectrometry in daily life.

(01 score)

Item 2:

Tororo Minerals Limited has developed a new inorganic fertilizer for maize labeled **Q** in a hydrated Form. Laboratory analysis shows that the solid fertilizer contains 9.37% magnesium, 10.93% nitrogen, and 42.18% water of crystallization and rest being oxygen. Molar mass of **Q** is 256gmol^{-1} .

The fertilizer was available for sale, the manager sought to know the percentage composition of the nitrate nutrient in **Q** that would be delivered into maize. Keen to know about environmental hazards, **Q** was heated strongly and decomposed with brown gas of nitrogen dioxide according to equation:

Equation 1: $2\text{Mg}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{MgO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$

The company sought to know the volume at r.t.p of the brown gas produced from burning 0.5kg of the fertilizer and assess the environmental impacts of the emissions and mitigations.

Due to available rocks containing magnesium in the area, the company manager further seeks to evaluate the industrial synthesis of the fertilizer **Q** from magnesium oxide:

Equation 2: $\text{MgO(s)} + 2\text{HNO}_3\text{(aq)} \rightarrow \text{Mg(NO}_3)_2\text{(aq)} + \text{H}_2\text{O(l)}$

The supplier reported that he would potentially supply 15.0 kg of magnesium oxide per day, and the manager intends to know how much fertilizer would be available in one month and also possible environmental impacts of using the fertilizer and the mitigations.

Task: As a learner of chemistry, help the company know;

(a) The Molecular formula of compound Q.

(06 scores)

(b) Percentage composition of nitrate nutrient in Q that would be delivered to maize. (03 scores)

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(ii) The environmental impacts of the brown gas emissions and mitigations. (02 scores)

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(d)(i) Mass of the fertilizer produced in one month. (04 scores)

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(ii) Possible environmental impacts of over using and poor management of the fertilizer. (02 scores)

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SECTION B

Part I (Inorganic Chemistry)

Attempt the item from this part on a separate answer booklet

Item 3:

At Green Valley Senior Secondary School in Kampala, a Senior Five student named Teddy had excelled in her Uganda Certificate of Education (UCE), scoring grade A in Chemistry. Confident in her abilities, she expected a smooth transition into A-level. However, during her first term studying Advanced Level Chemistry, she began to struggle.

During lessons on atomic structure, her teacher introduced new concepts such as **principal quantum shells, sub-energy levels (s, p, d, f), and orbitals**, which seemed different from the simpler shell model she had learned before. The teacher also emphasized that understanding electron arrangement requires applying Pauli Exclusion Principle, Aufbau Principle, and Hund's Rule.

Additionally, Teddy was puzzled when the teacher explained that some electron configurations are more stable than others and help to understand why some elements are more reactive than others. Feeling confused, Teddy approaches you, a fellow learner, for help in making sense of these ideas using practical examples and clear explanations.

Task: As a competent chemistry learner:

- (a) Help Teddy to understand how electrons are arranged in an atom by explaining the relationship between **principal quantum shells, sub-energy levels, and orbitals**. **(03 scores)**
- (b) Guide Teddy in understanding the principles and rule used when writing electron configurations by clearly explaining;
 - (i) Aufbau Principle, Pauli Exclusion Principle and Hund's Rule. Illustrate each using nitrogen as the suitable example. **(06 scores)**
 - (ii) Explaining why some configurations are more stable. Illustrate using copper and chromium as suitable examples. **(06 scores)**
- (c) Assist Teddy to correctly write the electron configurations of the following elements and ions; Aluminium, Phosphide ion, Nickel and Vanadium(III) ion. **(04 scores)**
- (d) Teddy wonders why this topic matters. Help her by explaining how electron configuration influences trend in chemical reactivity of;
 - (i) Alkaline earth metals (Group II elements).
 - (ii) Halogens (Group VII elements).
 - (iii) Noble gases (Group VIII elements). **(06 scores)**

Part II (Organic Chemistry)

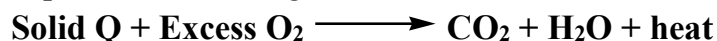
Attempt the item from this part on a separate answer booklet

Item 4:

A pharmaceutical plant in Kawempe division produces aspirin from **an organic acidic crystalline solid Q** and also produces paracetamol from **an organic crystalline solid P** which is amphoteric (*has both basic properties and acidic properties*).

Due to poor waste management policies in Uganda, excess and expired quantities of these raw materials are sometimes damped in nearby water bodies or disposed of by open-air burning at a local waste site on two different days. The following have been observed by the communities around the plant.

- ❖ On Mondays when excess or expired **solid Q** is burnt, residents complain of dense smoke, soot deposition, breathing discomfort and heat discomfort. **Q** burns according to equation:



- ❖ On Fridays when excess or expired **solid P** is burnt residents report eye irritation, choking brown fumes that cause smog and occasional acidic rain especially when it rains on Fridays. Substance **P** burns according to equation:



These observations have raised serious concerns about the health and environmental dangers associated with the poor methods of disposal, sparking off scientific investigations.

A team of chemists from NEMA came to investigate which raw material poses a greater risk to health and environment. They decided to separately burn samples of Q and P in excess oxygen and products collected and analyzed. The team presented the following experimental data about each solid.

Solid	Q	P
Mass of sample used	1.38g	5.45g
Mass of carbon dioxide produced by sample used	3.08g	13.21g
Mass of water produced by sample used	0.54g	3.15g
Percentage of nitrogen	None	12.84%
Vapour density of vaporized sample at s.t.p	6.161gdm ⁻³	4.8661gdm ⁻³

The chemists also obtained the following information from chemistry data books:

- Molar gas volume at s.t.p = 22.4dm³ mol⁻¹.
- Atomic masses: C=12, H=1, O=16, N=14.

Task: As a knowledgeable chemistry student, use the data above to assist the chemists to;

(a) Calculate the percentage composition of solid;

(i) Q. (02 scores)

(ii) P. (02 scores)

(b) Use the compositions obtained to find out the empirical formula of solid;

(i) Q. (03 scores)

(ii) P. (03 scores)

(c) Determine the molar mass and hence the molecular formula of solid;

(i) Q. (03 scores)

(ii) P. (03 scores)

(d) Use the results obtained in (c) above and observations given in scenario to explain why different pollution effects are experienced on different days. Hence state with a reason which substance poses greater risk to climate and which one poses greater risk to health. (03 scores)

(e) Suggest any other environmental danger of poor disposal of the excess or expired raw materials and hence suggest a mitigation measure. (01 score)

END